

Enterobacteriaceae, Coliforms, and *Escherichia Coli*

T Sandle, Bio Products Laboratory Ltd, Elstree, UK

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Introduction

The Enterobacteriaceae is a large family of Gram-negative bacteria that includes, along with many harmless symbionts, many of the more familiar pathogens, such as *Salmonella*, *Escherichia coli*, *Yersinia pestis*, *Klebsiella*, *Shigella*, *Proteus*, *Enterobacter*, *Serratia*, and *Citrobacter*. Members of the Enterobacteriaceae are rod shaped and typically are 1–5 µm in length. Enterobacteria have Gram-negative stains, and they are facultative anaerobes, fermenting sugars to produce lactic acid and various other end products. Many members of this family are a normal part of the gut flora found in the intestines of humans and other animals, whereas others are found in water or soil, or are parasites on a variety of different animals and plants (Williams et al., 2010).

Some bacteria grouped as Enterobacteriaceae are sub-categorized as ‘coliforms,’ including *Escherichia*, *Enterobacter*, *Klebsiella*, *Serratia*, and *Citrobacter*, because they share similar morphological and biochemical characteristics. Coliform bacteria are a commonly used bacterial indicator of sanitary quality of foods and water (Taylor, 1986).

Given the importance of these bacteria, as food-poisoning organisms, the accurate identification of the different species is of great importance to food microbiology. The identification of microorganisms with conventional methods involves a great amount of materials and work. Therefore, many rapid identification systems have been developed (MacFaddin, 1980). These techniques are based on ready-to-use media, or more accurate methods using dehydrated substrates, placed in cupules or tubes, or rapid microbiological methods (Kilian and Bulow, 1976).

There are a number of conventional and commercially available automated and nonautomated systems to identify Gram-negative rods. Except for reference testing, conventional macrotube biochemical tests for bacterial identification have been replaced by commercial systems, because the classical methods are too expensive, slow, and unwieldy for routine use in the microbiological laboratory. Some of the systems are restricted to one genus (i.e., API Listeria), and others can be applied for large groups (BBL Crystal Grampositive Identification System). These systems range from visual interpretation of miniaturized biochemical panels with computerized taxonomic databases to semiautomated or automated systems that can interpret and analyze results in a matter of hours. Many laboratories now adopt semiautomated phenotypic identification systems, such as VITEK or Omnilog, or they have embraced genotypic methods, including polymerase chain reaction (PCR)-based methods.

Systems for determination of Enterobacteriaceae are more available than identification systems for other microorganisms

(Griffioen and Beumer, 1995). As most of these systems initially were developed for application in clinical microbiology, bacteria from animal, food, feed, or environmental sources may be tested less commonly and incorporated in the identification databases of the system. The diversity of these microorganisms, originating from various sources, may cause problems for the identification systems, since bacterial strains of the same species may vary slightly in their biochemical reactions. There are also differences among the databases (Phenetic Classification Database) of different identification systems.

Although results obtained with various identification systems for Enterobacteriaceae have been described in the literature, worldwide most tests for the biochemical identification of Enterobacteriaceae (in medical, industrial, and research laboratories) are performed with the miniaturized systems API 20E, BBL® Enterotube™, and BBL® Crystal™ Enteric/Nonfermenter ID. This article described these systems in detail, and the results obtained with the systems in identifying test strains of Enterobacteriaceae are discussed.

Principles and Types of Commercially Available Tests

Identification methods can be divided into two groups: phenotypic and genotypic. The genotype–phenotype distinction is drawn in genetics. ‘Genotype’ is an organism’s full hereditary information, even if not expressed. ‘Phenotype’ is an organism’s actual observed properties, such as morphology, development, or behavior. Phenotypic methods are the most widespread due to their relatively lower costs for many laboratories. Expressions of the microbial phenotype – that is, cell size and shape, cellular composition, antigenicity, biochemical activity, sensitivity to antimicrobial agents, and so on – frequently depend on the media and growth conditions that have been used.

Phenotypic Methods

Phenotypic methods tend to work on the process of elimination. If test A is positive and B is not, then one group of possible microorganisms is included and another is excluded. From this, tests C and D are performed, and so on. The test results are compared against databases that work on the basis of a dichotomous key. A dichotomous key is a way of dividing groups of organisms, based on certain attributes. Bacteria are categorized using this method based on differences in their physical or metabolic attributes.

Phenotypic methods can be divided into manual and semiautomated methods.

Manual Test Kits

Once the cellular characteristics of the unknown organism have been reported (such as through a microscopic view of the Gram reaction), the second stage of identification is to identify the genus and species of bacteria. The most common techniques used, based on their costs and long history, are biochemical tests. Before starting a test, it must be confirmed that the culture is an Enterobacteriaceae. To test this, specific growth on violet red bile agar, fermentation of glucose (positive), and oxidase reaction (negative) should be observed (Sutton and Cundell, 2004). The oxidase test is for cytochrome *c* oxidase. Enterobacteriaceae are typically oxidase negative, meaning they either do not use oxygen as an electron acceptor in the electron transport chain, or they use a different cytochrome enzyme to transfer electrons to oxygen. If the culture is determined to be oxidase positive, alternative tests must be carried out to correctly identify the bacterial species. The subsequent Gram stain should be negative, and the morphology of the stained cells should be rod shaped (Hartman, 1968).

Phenotypic reactions typically incorporate reactions to different chemicals or different biochemical markers. These rely on the more subjective determinations. The reliance on biochemical reactions and carbon utilization patterns introduces some disadvantages to the achievement of consistent (repeatable and reproducible) identification. These are mature technologies, however, such as the API strip, that are marketed by multiple companies as consistent, prepackaged kits with well-established quality control procedures (Monnet et al., 1994).

For the Enterobacteriaceae, the appropriate test kits are the API 20E (bioMérieux, Marcy-l'Etoile, France), BBL® Enterotube™, and BBL® Crystal™ (Becton Dickinson and Company, Maryland, United States). These test kits do not require high investments in apparatus, are user friendly, and are well known in laboratories for clinical, veterinary, and food microbiology (Mickewright and Sartory, 1995).

Principle and Use of the API 20E Identification System for Enterobacteriaceae

API 20E is a standardized identification system for Enterobacteriaceae and other nonfastidious Gram-negative rods, which uses 23 miniaturized biochemical tests and a database. The API 20E strip consists of 20 microtubes containing dehydrated substrates. These tests are inoculated with a bacterial suspension that reconstitutes the media. During incubation, metabolism or metabolite produces color changes that either are spontaneous or revealed by the addition of reagents. The obtained reactions are analyzed according to the interpretation table, the analytical profile index (a codebook), or the APILAB software (or, more commonly, an online resource).

The API 20E kit consists of 25 API 20E strips and 25 incubation boxes. To use the API 20E, the following materials are necessary: suspension medium or sterile water (5 ml), reagent kit, zinc reagent, mineral oil, pipettes, API 20E profile index or identification software, and an ampoule

rack. The following general laboratory equipment also is required: incubator (35–37 °C), refrigerator, Bunsen burner, and marker pen.

To determine fermentative or oxidative metabolism and motility, API OF-medium and API M-medium might be necessary.

Directions for Use

Preparation of the Strip

- Prepare an incubation box, tray, and lid and distribute about 5 ml of water into the honeycombed wells of the tray to create a humid chamber.
- Record the strain reference on the elongated tab of the tray.
- Place the strip in the tray.
- Perform the oxidase test on an identical colony. Only use oxidase-negative colonies for the biochemical determination of Enterobacteriaceae.

Preparation of the Inoculum

- Open an ampoule of suspension medium or sterile water without additives.
- With the aid of a pipette, remove a single well-isolated colony from an isolation plate.
- Carefully emulsify to achieve a homogenous bacterial suspension.

Inoculation of the Strip

- With the same pipette, fill both the tube and cupule of tests CIT (citrate utilization), VP (acetoin production, Voges-Proskauer reaction), and GEL (gelatinase), with the bacterial suspension.
- Fill only the tubes (and not the cupules) of the other tests.
- Create anaerobiosis in the tests ADH (arginine dihydrolase), LDC (lysine decarboxylase), ODC (ornithine decarboxylase), URE (urease), and H₂S by overlaying with mineral oil.
- Close the incubation box and incubate at 35–37 °C for 18–24 h.

Reading of the Strip

- After 18–24 h at 35–37 °C, read the strip by referring to the interpretation table.
- Record all spontaneous reactions on the record sheet.
- If the glucose reaction is positive or three tests or more are positive, reveal the results that require the addition of reagents: VP, TDA (tryptophane desaminase), IND (indole production), and NO₂.
- Add the reagents required and record the results on the report sheet.
- If the glucose reaction is negative and the number of positive tests is less than or equal to two, do not add reagents.

Identification

- Using the identification table, compare the results recorded on the report sheet with those given in the table.

- With the analytical profile index or the identification software, code the pattern of the reactions obtained in a numerical profile.

On the report sheet, the tests are separated into groups of three, and a number (4, 2, or 1) is indicated for each. By adding the numbers corresponding to positive reactions within each group, a seven-digit profile number is obtained for the 20 tests of the API 20E strip.

In some cases, the seven-digit profile is not discriminatory enough, and additional tests should be carried out, including the following: reduction of nitrates to nitrites (NO_2), reduction of nitrites to N_2 gas, motility, growth on MacConkey agar medium, oxidation of glucose, and fermentation of glucose.

Principle of the BBL® Enterotube™ for the Identification of Enterobacteriaceae

The Enterotube™ and its computer coding and identification system are designed specially for the identification of Enterobacteriaceae (i.e., aerobic, Gram-negative rods, which are oxidase negative). The Enterotube™ consists of 12 compartments in a row, filled with ready-to-use media ('wet' system), with which 15 reactions can be performed.

Enterotubes™ are delivered in boxes containing 5 or 25 units, 5 or 25 report sheets, and an instruction manual.

Directions for Use

Preparation of the Tube

Remove both caps from the tube. The tip of the inoculation needle is under the white cap. Without flaming the needle, pick a well-isolated colony directly on the top of the needle. Do not puncture the agar.

Inoculation of the Tube

- Inoculate the tube by first twisting the needle, then withdrawing it through all the compartments using a turning motion. Reinsert the needle (without sterilizing) into the tube until the notch on the needle is aligned with the opening of the tube. The tip of the needle should be seen in the citrate compartment.
- Break the needle at the notch by bending. The portion of the needle remaining in the tube maintains the anaerobic conditions necessary for true fermentation of glucose, decarboxylation of lysine and ornithine, and the detection of gas production. Replace both caps.
- With the broken-off part of the needle, punch holes through the foil, covering the air inlets of the last eight compartments (adonitol, lactose, arabinose, sorbitol, Voges-Proskauer, and dulcitol/PA (phenylalanine desaminase), urea, and citrate) to support aerobic growth in these compartments.
- Incubate the tube at 35–37 °C, standing it upright if possible in the test-tube support with its glucose compartment pointing upward, or lay the tube on its flat surface.

Reading of the Tube

- Interpret all the reactions with the exception of indole and Voges-Proskauer, in comparison with a reference picture or a not-inoculated tube. Record the reactions on the interpretation pad. The reaction is negative if the compartment remains unchanged (exceptions: indole and Voges-Proskauer).
- Perform the indole test and the Voges-Proskauer test by adding the Kovacs's reagent into the H₂S/indole compartment (directly under the plastic film) and α -naphthol and potassium hydroxide (through the air inlet) into the Voges-Proskauer compartment.

To identify an isolate, the numbers for the positive reactions are written down on the interpretation pad. The circled numbers for 5 × 3 reactions are added together, resulting in a five-digit number (ID value). The number thus obtained then is compared with the Computer Coding and Identification system (available at Becton Dickinson), which results in the identification of a microorganism.

If further tests are required, or if the purity of culture has to be checked, an inoculum from the incubated tube can be taken and applied to a suitable medium or broth for subcultivation as follows:

- The inoculation needle is drawn out with sterile forceps and streaked on a plate.
- Bacterial substance is extracted from a positive compartment with a sterile loop after the plastic film has been removed.

Principle of the BBL® Crystal™ Enteric/Nonfermenter ID System for Enterobacteriaceae

The BBL® Crystal™ Enteric/Nonfermenter (E/NF) identification system is a miniaturized identification method employing modified conventional and chromogenic substrates. The kit includes lids, bases, and inoculum fluid tubes. The lid contains 30 dehydrated substrates on the tips of plastic prongs. The base has 30 reaction wells. The test inoculum is prepared with the inoculum fluid and is used to fill all 30 wells in the base. When the lid is aligned with the base and snapped in place, the test inoculum rehydrates the dried substrates and this rehydration initiates test reactions.

After an incubation period, the wells are examined for color changes, resulting from metabolic activity of microorganisms. The resulting pattern of the 30 reactions is converted into a 10-digit profile number that is the basis for identification.

The BBL® Crystal™ kit consists of 20 lids, 20 bases, 20 tubes with inoculum fluid, two incubation trays, one color reaction card, and a results pad. To use the kit, the following materials are required (but not provided): sterile cotton swabs, incubator (35–37 °C, 40–50% humidity), a BBL® Crystal™ light box, the BBL Crystal™ ID system electronic codebook, nonselective culture plates (e.g., tryptone-soy agar), and reagents to perform the indole test and the oxidase test (BBL DMACA Indole and BBL Oxidase reagent dropper) (Holmes et al., 1994).

Directions for Use

Preparation of the Panels

- Remove the lids from the pouch. Discard the dust cover and desiccant. Once removed from the pouch, the lids should be used within 1 h.
- Take a tube with inoculum fluid and label it with the number of the strain to be tested. Using an aseptic technique, pick one well-isolated large (~2–3 mm) colony (or 4–5 smaller colonies of the same morphology) with the tip of a sterile cotton swab or a wooden applicator stick from a blood plate such as Trypticase[®] soy agar with 5% sheep blood or MacConkey agar (or any suitable nonselective medium).
- Suspend the colony material in the inoculum fluid.
- Recap the tube and vortex for approximately 10–15 s.
- Take a base, and mark the number of the strain on the side wall.
- Pour the entire contents of the inoculum into the target area of the base.
- Hold the base in both hands and roll the inoculum gently along the tracks until all of the wells are filled. Roll back any excess fluid to the target area and place the base on a bench top.
- Align the lid so that the labeled end of the lid is on top of the target area of the base.
- Push down until a slight resistance is felt. Place your thumb on the edge of the lid toward the middle of panel on each side and push downward simultaneously until the lid snaps into place (listen for two 'clicks').
- Place inoculated panels in incubation trays. All panels should be incubated upside-down in a non-CO₂ incubator with 40–60% humidity. Trays should not be stacked more than two high during incubation. The incubation time is 18–20 h at 35–37 °C.

Reading of the Panels

- After the recommended period of incubation, remove the panels from the incubator. All panels should be read upside-down using the BBL[®] Crystal[™] light box. Refer to the color reaction chart for an interpretation of the reactions. Use the BBL[®] Crystal E/NF results pad to record reactions.
- Each test that is positive is given a value of 4, 2, or 1, corresponding to the row at which the test is located. A value of 0 (zero) is given to any negative result. The numbers resulting from each positive reaction in each column are then added together, resulting in a 10-digit number: the BBL[®] Crystal[™] profile number.
- This number, and the offline tests for indole and oxidase tests, should be entered on a personal computer in which the BBL[®] Crystal[™] ID system electronic codebook has been installed to obtain the identification.

Comparison of the Manual Tests

The biochemical reactions used in the identification systems can be classified according to the reaction patterns in the following groups: decarboxylation reactions, hydrolysis,

oxidation or fermentation reactions, production of characteristic substances, and 'other reactions.' The biochemical reactions used for each of the three identification systems are given in Table 1.

The criterion for a 'good identification' is 90% confidence (Holmes et al., 1977). A percentage level is provided by the BBL[®] Crystal[™] system and the BBL[®] Enterotube[™] (78%). With the API 20E test, in addition to the confidence percentage, a so-called T-index is provided. This value varies from 0 to 1 and is inversely proportional to the number of atypical reactions. According to the manufacturer, apart from a percentage of identification of at least 90%, percentages below 90% with a T-index between 0.5 and 1.0 are also acceptable.

Problems can arise in cases in which an inadequate identification is obtained. This can be due to a short incubation time or a too low level of inoculation.

Semiautomated Methods

Some of the commercially available methods – for example, VITEK (bioMérieux, Marcy-l'Etoile, France) or Omnilog (Biolog Inc., Hayward, California) – can be used only in combination with expensive equipment. Nonetheless, such systems reduce costs when processing a large volume of samples (Stager and Davis, 1992). Both systems are growth-based, biochemical and carbohydrate utilization.

With the VITEK system, microbial cells from isolated colonies are used to prepare a microbial suspension, which then is added to specific test cards containing substrates for enzymatic utilization, carbohydrate acidification, and other tests. Color or turbidity changes in each well are measured every 15 min and results are compared with an internal library. Gram staining is required to determine the correct test card to use (Gherardi et al., 2012).

With the Omnilog, microbial cells from isolated colonies are used to prepare a microbial suspension, which then is added to specific test cards containing a variety of carbohydrates and a colorless tetrazolium violet dye (Klingler et al., 1992). If growth occurs, the dye turns violet in color. The resulting color patterns are compared with an internal library (Shea et al., 2012).

Alternative rapid, automated phenotypic systems include the Hy-enterotest (Hy laboratories, Israel), Phoenix (BD Diagnostics, United States), MALDI Biotyper (Bruker Daltonics), and the Sherlock MIS (MIDI) (Miller, 2012).

Genotypic Methods

Genotypic methods are not reliant on the isolation medium or growth characteristics of the microorganism. Genotypic methods have considerably enhanced databases of different types of microorganisms. In contrast to the phenotypic methods, genotypic techniques are more accurate. This is because the microbial genotype is highly conserved and is independent of the culture conditions, so the identifications may be conducted on uncultured test material—primary enrichments that increase the amount of nucleic acid available for analysis (Dutka-Malen et al., 1995).

Table 1 Biochemical reactions used in three identification systems for Enterobacteriaceae (API 20E, BBL® Enterotube™, and BBL® Crystal™)

Type of reaction	API 20E	Enterotube™	BBL® Crystal™
Decarboxylation	Lysine, ornithine	Lysine, ornithine	Lysine
Hydrolysis	Arginine o-n-p (ortho-nitro-phenyl) galactoside, urea	Urea	Arginine, aesculin, p-n-p-acetyl-glucosaminide, p-n-p-arabinoside, p-n-p-bisphosphate, p-n-p-β-galactoside, p-n-p-α-β-glucoside, p-n-p-β-glucuronide, p-n-p-γ-L-glutamyl-p-n-anilide, p-n-p-phosphate, p-n-p-phosphorylcholine, p-n-p-xyloside, proline p-n-anilide, urea
Oxidation	Amygdalin, arabinose, citrate, glucose, inositol, mannitol, melibiose, rhamnose, sorbitol, sucrose	Adonitol, arabinose, citrate, dulcitol, glucose, lactose, sorbitol	Arabinose, adonitol, citrate, galactose, inositol, malonate, mannitol, mannose, melibiose, rhamnose, sorbitol, sucrose
fermentation			
Production	Acetoin, H ₂ S, indole, nitrite	Gas (glucose), H ₂ S, pyruvic acid, indole	–
Other reactions	Gelatin (degradation), tryptophan (deamination)	–	Tetrazolium (reduction), p-n-DL-p-alanine (oxidative deamination), glycine (degradation)

Genotypic microbial identification methods based on nucleic acid analyses are less subjective, less dependent on the culture method, and theoretically more reliable because nucleic acid sequences are highly conserved by microbial species. These methods would include DNA–DNA hybridization, PCR, 16s and 23s rRNA gene sequencing (the 16s rRNA gene is most commonly used), and analytical ribotyping (Kolbert and Persing, 1999). Given the expense of genotypic systems and the current low use in food microbiology laboratories, they are not discussed any further.

Conclusion

This article has examined the main methods for the identification of bacteria collectively grouped as Enterobacteriaceae, Coliforms, and *E. coli*. Given that most food microbiology laboratories continue to use manual identification test kits, the emphasis is on these techniques. Reference has been made, however, to semiautomated methods and to genotyping.

See also: Biochemical and Modern Identification Techniques: Introduction; Biochemical and Modern Identification Techniques: Food-Poisoning Microorganisms; Enterobacteriaceae, Coliform, and *Escherichia coli*: Classical and Modern Methods for Detection and Enumeration; Enzyme Immunoassays: Overview; Hydrophobic Grid Membrane Filter Techniques; Immunomagnetic Particle-Based Techniques: Overview; National Legislation, Guidelines, and Standards Governing Microbiology: European Union; National Legislation, Guidelines, and Standards Governing Microbiology: Japan; Nucleic Acid–Based Assays: Overview; Petrifilm– A Simplified Cultural Technique; Rapid Methods for Food Hygiene Inspection; Sampling Plans on Microbiological Criteria; Water Quality Assessment: Modern Microbiological Techniques.

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